Supplementary material for "Do neural and zero-shot models beat persistence in stock price forecasting? A cross-market study"

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1 Purpose and scope

This supplement reports the corrected experiment and the scope of its verification. The main manuscript presents the central result, while the tables here retain list, horizon, feature, and seed distinctions. The experiment includes 35,640 supervised rows, 990 sampled-origin Chronos rows, and 60 full-origin subset rows. These rows form a fixed grid with shared assets and related configurations, not independent trials. The corrected local batch is `full_corrected_20261109`; the earlier public archive contains a different evaluation version.

2 Results and evidence files

Tables 1 and 2 separate supervised and sampled-origin results by asset list. Table 3 reports the small full-origin subset, which contains nine unique tickers. Tables 4 and 5 retain the price mode, feature set, horizon, and seed for selected supervised rows. All win counts use strict comparisons at full precision, so a displayed ratio rounded to 1.00 does not determine a gate result. The underlying files retain full numeric precision and the complete experiment keys.

Table 1: Supervised results by list.

List	Implementation	Rows	RMSE wins	All-three	Median
Dow 30	GRU	1,080	5	5	1.98
Dow 30	LSTM	1,080	2	1	2.44
Dow 30	Linear temporal proxy	1,080	0	0	1.92
Dow 30	PatchTST-style	1,080	0	0	3.49
Dow 30	TimeMixer-style	1,080	2	1	2.20
Dow 30	iTransformer-style	1,080	0	0	3.95
ETF basket	GRU	432	5	3	3.58
ETF basket	LSTM	432	0	0	4.56
ETF basket	Linear temporal proxy	432	0	0	2.19
ETF basket	PatchTST-style	432	0	0	5.71
ETF basket	TimeMixer-style	432	0	0	3.12
ETF basket	iTransformer-style	432	0	0	6.51
Hang Seng large-cap	GRU	1,548	6	0	1.30
Hang Seng large-cap	LSTM	1,548	2	0	1.47
Hang Seng large-cap	Linear temporal proxy	1,548	0	0	2.67
Hang Seng large-cap	PatchTST-style	1,548	2	1	1.81
Hang Seng large-cap	TimeMixer-style	1,548	3	1	1.81
Hang Seng large-cap	iTransformer-style	1,548	3	1	1.85
Nasdaq large-tech	GRU	1,080	0	0	3.67
Nasdaq large-tech	LSTM	1,080	0	0	3.98
Nasdaq large-tech	Linear temporal proxy	1,080	0	0	1.86
Nasdaq large-tech	PatchTST-style	1,080	0	0	5.10
Nasdaq large-tech	TimeMixer-style	1,080	0	0	2.44
Nasdaq large-tech	iTransformer-style	1,080	0	0	5.57
S&P 100 subset	GRU	1,800	5	5	2.09
S&P 100 subset	LSTM	1,800	2	1	2.48
S&P 100 subset	Linear temporal proxy	1,800	0	0	2.21
S&P 100 subset	PatchTST-style	1,800	0	0	3.78
S&P 100 subset	TimeMixer-style	1,800	2	1	2.35
S&P 100 subset	iTransformer-style	1,800	0	0	4.13

Counts use full precision; ratios use two decimals. Entries from overlapping lists and related configurations are dependent.

Table 2: Sampled-origin Chronos results by list.

List	Implementation	Rows	RMSE wins	All-three	Median
Dow 30	Chronos sampled	180	13	9	1.03
ETF basket	Chronos sampled	72	12	10	1.05
Hang Seng large-cap	Chronos sampled	258	41	22	1.04
Nasdaq large-tech	Chronos sampled	180	14	7	1.04
S&P 100 subset	Chronos sampled	300	39	19	1.03

Counts use full precision; ratios use two decimals. Entries from overlapping lists and related configurations are dependent.

Table 3: Full-origin Chronos subset by list.

List	Rows	RMSE wins	All-three	Median
Dow 30	12	0	0	1.03
ETF basket	12	1	1	1.04
Hang Seng large-cap	12	0	0	1.03
Nasdaq large-tech	12	0	0	1.03
S&P 100 subset	12	0	0	1.03

The subset contains ten asset entries but nine unique tickers, selected as the first two entries per list.

Table 4: Ten lowest supervised RMSE ratios.

Dataset	Ticker	Implementation	Days	Price	Features	Seed	Ratio
Hang Seng	9618.HK	Inverted style	5	Raw	OHLCV	42	0.94
Dow	DIS	GRU	10	Raw	Indicators	123	0.94
S&P 100	DIS	GRU	10	Raw	Indicators	123	0.94
ETF	TLT	GRU	10	Adj	OHLCV	777	0.96
Hang Seng	2020.HK	Mixing style	10	Raw	Indicators	777	0.96
Hang Seng	9618.HK	Inverted style	5	Raw	OHLCV	123	0.97
Dow	DIS	GRU	10	Adj	Indicators	42	0.97
S&P 100	DIS	GRU	10	Adj	Indicators	42	0.97
S&P 100	DIS	GRU	10	Raw	OHLCV	42	0.97
Dow	DIS	GRU	10	Raw	OHLCV	42	0.97

These are selected supervised rows, not separate asset-level discoveries. The fixed window is 60. Raw and Adj identify downloaded-close and adjusted-close modes; Linear proxy, Inverted style, and Mixing style denote the linear temporal proxy, iTransformer-style, and TimeMixer-style. Indicators means OHLCV plus technical indicators.

Table 5: Ten highest supervised RMSE ratios.

Dataset	Ticker	Implementation	Days	Price	Features	Seed	Ratio
Hang Seng	1044.HK	Mixing style	1	Raw	OHLCV	42	60.03
Hang Seng	1044.HK	Mixing style	1	Raw	OHLCV	777	52.36
ETF	HYG	Linear proxy	1	Adj	OHLCV	123	47.39
Hang Seng	1044.HK	Mixing style	1	Raw	OHLCV	123	45.69
Hang Seng	1044.HK	Mixing style	1	Adj	OHLCV	42	43.09
S&P 100	BK	Linear proxy	1	Raw	Indicators	777	38.29
S&P 100	BK	Linear proxy	1	Raw	OHLCV	777	36.56
S&P 100	BK	Linear proxy	1	Raw	Indicators	123	36.36
Hang Seng	0968.HK	Linear proxy	1	Raw	OHLCV	42	34.95
Hang Seng	9618.HK	Linear proxy	1	Raw	OHLCV	42	34.52

These are selected supervised rows, not separate asset-level discoveries. The fixed window is 60. Raw and Adj identify downloaded-close and adjusted-close modes; Linear proxy, Inverted style, and Mixing style denote the linear temporal proxy, iTransformer-style, and TimeMixer-style. Indicators means OHLCV plus technical indicators.

The local evidence directories `corrected_evidence` and `full_origin_evidence` contain separate combined results and model, horizon, and list summaries. Per-run files retain input and source hashes, arguments, package versions, dated predictions, and verification outputs. The supervised batch also stores fitted checkpoints and feature scalers. The project preserves the old archive, so its original numbers remain traceable rather than silently overwritten. A later public release must identify the corrected batch separately.

3 Actual time splits

The split rule assigns 72%, 8%, and 20% of each condition's usable observations to training, validation, and testing before boundary removal. Training and validation labels must have target dates strictly before the next segment's first forecast origin. Table 6 shows the minimum and maximum first-origin dates across assets and price modes for each list, feature set, and horizon. A range denotes variation across conditions, not an interval used as one shared split. The complete file `split_boundary_exact.csv` records dates, retained sample counts, and removed boundary rows for all 1,980 conditions.

Table 6: Ranges of actual split origins.

List	Features	Days	First validation origin	First test origin
Dow 30	OHLCV	1	23-02-2023	22-01-2024
Dow 30	OHLCV	5	17-02-2023	17-01-2024
Dow 30	OHLCV	10	14-02-2023	10-01-2024
Dow 30	Indicators	1	09-03-2023	31-01-2024
Dow 30	Indicators	5	06-03-2023	26-01-2024
Dow 30	Indicators	10	28-02-2023	22-01-2024
ETF basket	OHLCV	1	23-02-2023	22-01-2024
ETF basket	OHLCV	5	17-02-2023	17-01-2024
ETF basket	OHLCV	10	14-02-2023	10-01-2024
ETF basket	Indicators	1	09-03-2023	31-01-2024
ETF basket	Indicators	5	06-03-2023	26-01-2024
ETF basket	Indicators	10	28-02-2023	22-01-2024
Hang Seng large-cap	OHLCV	1	21-02-2023 to 03-09-2024	18-01-2024 to 26-02-2025
Hang Seng large-cap	OHLCV	5	16-02-2023 to 29-08-2024	15-01-2024 to 20-02-2025
Hang Seng large-cap	OHLCV	10	13-02-2023 to 26-08-2024	09-01-2024 to 14-02-2025
Hang Seng large-cap	Indicators	1	06-03-2023 to 19-09-2024	29-01-2024 to 06-03-2025
Hang Seng large-cap	Indicators	5	02-03-2023 to 13-09-2024	24-01-2024 to 03-03-2025
Hang Seng large-cap	Indicators	10	24-02-2023 to 09-09-2024	18-01-2024 to 25-02-2025
Nasdaq large-tech	OHLCV	1	23-02-2023	22-01-2024
Nasdaq large-tech	OHLCV	5	17-02-2023	17-01-2024
Nasdaq large-tech	OHLCV	10	14-02-2023	10-01-2024
Nasdaq large-tech	Indicators	1	09-03-2023	31-01-2024
Nasdaq large-tech	Indicators	5	06-03-2023	26-01-2024
Nasdaq large-tech	Indicators	10	28-02-2023	22-01-2024
S&P 100 subset	OHLCV	1	23-02-2023	22-01-2024
S&P 100 subset	OHLCV	5	17-02-2023	17-01-2024
S&P 100 subset	OHLCV	10	14-02-2023	10-01-2024

List	Features	Days	First validation origin	First test origin
S&P 100 subset	Indicators	1	09-03-2023	31-01-2024
S&P 100 subset	Indicators	5	06-03-2023	26-01-2024
S&P 100 subset	Indicators	10	28-02-2023	22-01-2024

4 Verification and interpretation

The strict verification covers the complete experiment-key sets and checks for duplicates. It reconstructs the declared test origins, confirms target dates and prices against the hashed inputs, and recomputes model and persistence errors, ratios, and gates. An independent calculation checks 19,845,948 saved prediction points across the three protocols. Additional reconstruction confirms the training and validation boundaries, sample counts, and target normalization for all 1,980 supervised conditions. These calculations agree with the saved results and the generated aggregate summaries.

Checkpoint replay covers 72 selected cases across six implementations, two price modes, two feature sets, and three horizons, rotating across the five lists. The replay matches the saved inference outputs, configurations, and feature scalers. This is a stratified check, not an independent retraining or replay of every model. The manifests record a fixed Chronos revision, but the separately recorded cache-file hashes describe the inspected cache rather than a byte-level record captured at the original inference run. Neither arithmetic verification nor a zero-shot label proves historical adjustment-factor availability, a representative asset universe, or absence of pretraining-data overlap.

The earlier technical-feature branch compares 990 conditions, or 17,820 supervised rows, with baseline errors from different test dates. The corrected batch aligns every model with its own baseline and removes labels that reach the next evaluation segment. Other corrections and runtime differences also accompany the rerun, so changes from the old archive do not identify the effect of one repair in isolation. Cross-feature and cross-protocol comparisons still have different usable dates and inputs. We therefore use descriptive summaries and make no row-independent confidence-interval or significance claim.